

E2. Quartz Crystal Oscillator (Simulation) (10.0 points)

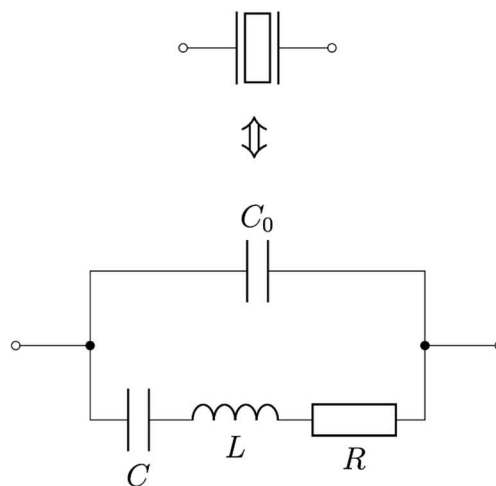
<https://taisoncar.github.io/quartz/>

Credit: Russian Team Selection Test (X) 2026 [https://pho.rs/p/4788]

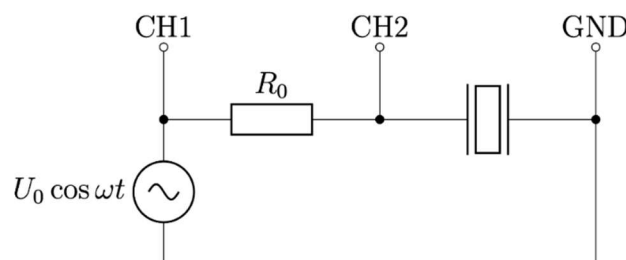
A quartz crystal oscillator is a type of electrical circuit that utilizes the piezoelectric properties of quartz crystal to generate electrical oscillations. Quartz crystal oscillator is widely used to keep track of time in watches and computers due to its stability and accuracy.



To analyze the behavior of a quartz crystal in electrical circuits, we can model it as an equivalent circuit shown in the figure below.



In this problem, we will investigate the parameters of a quartz crystal through a program that simulates the following circuit:



The resonant frequency of the quartz crystal used in this simulation is known to be close to $f_0 = 2 \text{ MHz}$. The circuit is powered by an AC source $U = U_0 \cos(\omega t)$ with angular frequency $\omega = 2\pi f$. The voltage amplitude of the AC source U_0 can be assumed to be constant at 500 mV . A resistor $R_0 = 5.1 \text{ k}\Omega$ is connected in series to the quartz crystal.

Note: Refer to the end of this paper for the instructions of the simulation program.

Part A. Measurements without external capacitor (5.7 points)

For this part, avoid connecting any external capacitor by inputting $C_{ext} = 0$.

- | | | |
|------------|--|--------|
| A1. | Determine the frequency f_{max} at which the voltage amplitude U_2 across the quartz crystal is maximized. | 0.2 pt |
|------------|--|--------|

$$f_{max} = 2000040 \text{ Hz}$$

- | | | |
|------------|--|--------|
| A2. | Determine the frequency f_{min} at which the voltage amplitude U_2 across the quartz crystal is minimized. | 0.1 pt |
|------------|--|--------|

$$f_{max} = 1999931 \text{ Hz}$$

- | | | |
|------------|---|--------|
| A3. | Measure the voltage amplitude U_2 across the quartz crystal and the phase difference φ between the signals on the two oscilloscope channels as frequency f changes. | 2.8 pt |
|------------|---|--------|

Just collect the data

- | | | |
|------------|--|--------|
| A4. | Express the magnitude of the impedance of the quartz crystal $ Z $ in terms of R_0 , U_2 , U_0 and φ . | 0.5 pt |
|------------|--|--------|

$$\frac{U_2 e^{i(\omega t + \varphi)}}{U_0 e^{i\omega t}} = \frac{Z}{R_0 + Z}$$

$$\frac{U_2}{U_0} e^{i\varphi} = \frac{Z}{R_0 + Z}$$

$$Z = \frac{R_0 e^{i\varphi}}{\frac{U_0}{U_2} - e^{i\varphi}}$$

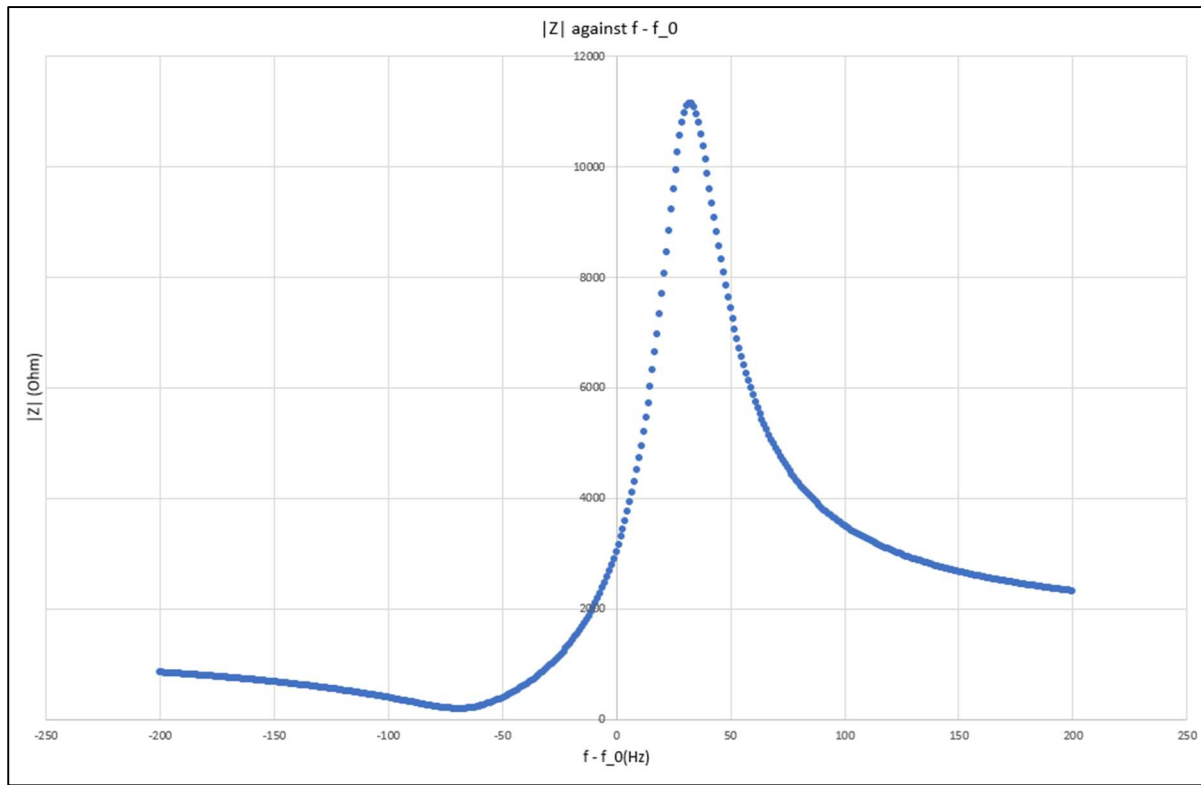
$$|Z| = \frac{|R_0 e^{i\varphi}|}{\left| \frac{U_0}{U_2} - e^{i\varphi} \right|}$$

$$|Z| = \frac{R_0}{\sqrt{\left(\frac{U_0}{U_2} - \cos \varphi \right)^2 + \sin^2 \varphi}}$$

A5.

Plot the magnitude of the impedance of the quartz crystal $|Z|$ against frequency f .

2.0 pt

**A6.**

From the plotted graph, find the resonant frequency f_{res} and anti-resonant frequency f_{anti} at which $|Z|$ reaches its maximum and minimum values respectively.

0.1 pt

$$f_{res} = 2000032 \text{ Hz}$$

$$f_{anti} = 1999933 \text{ Hz}$$

Part B. Deriving useful expressions (0.6 points)

The impedance of a quartz crystal can be written in the form

$$Z(\omega) = \frac{\omega_s^2 - \omega^2 + i\omega\alpha}{i\omega C_0(\omega_p^2 - \omega^2 + i\omega\alpha)}$$

where ω_s , ω_p and α do not depend on ω .

B1. Express ω_s , ω_p and α in terms of R , L , C and C_0 .

0.3 pt

$$\frac{1}{Z} = \frac{1}{\frac{1}{i\omega C} + i\omega L + R} + i\omega C_0$$

$$\frac{1}{Z} = \frac{1 + i\omega C_0 \left(\frac{1}{i\omega C} + i\omega L + R \right)}{\frac{1}{i\omega C} + i\omega L + R}$$

$$Z = \frac{\frac{1}{LC} - \omega^2 + \frac{i\omega R}{L}}{\frac{i\omega}{L} + i\omega C_0 \left(\frac{1}{LC} - \omega^2 + \frac{i\omega R}{L} \right)}$$

$$Z = \frac{\frac{1}{LC} - \omega^2 + \frac{i\omega R}{L}}{i\omega C_0 \left(\frac{1}{LC_0} + \frac{1}{LC} - \omega^2 + \frac{i\omega R}{L} \right)}$$

By comparison:

$$\omega_s = \frac{1}{\sqrt{LC}}; \quad \omega_p = \sqrt{\frac{1}{LC_0} + \frac{1}{LC}}; \quad \alpha = \frac{R}{L}$$

B2. Express $|Z|^2$ in terms of ω , ω_s , ω_p , α and C_0 .

0.1 pt

$$|Z|^2 = \frac{|\omega_s^2 - \omega^2 + i\omega\alpha|^2}{|\omega C_0(i(\omega_p^2 - \omega^2) - \omega\alpha)|^2}$$

$$|Z|^2 = \frac{(\omega_s^2 - \omega^2)^2 + (\omega\alpha)^2}{(\omega C_0)^2 \left((\omega_p^2 - \omega^2)^2 + (\omega\alpha)^2 \right)}$$

From here onwards, assume the following approximations hold for the range of frequencies we are working on in this problem:

$$Z(\omega) \approx \frac{\omega_s^2 - \omega^2 + i\omega_p\alpha}{i\omega_p C_0(\omega_p^2 - \omega^2 + i\omega_p\alpha)}$$

$$\alpha \ll \omega_p - \omega_s; \quad \alpha \ll \omega_p; \quad \alpha \ll \omega_s$$

$$\varphi(f_{anti}) = \varphi(f_{res}) = 0$$

B3. Show that $\omega_p = 2\pi f_{res}$ and $\omega_s = 2\pi f_{anti}$.

0.2 pt

$$\omega = 2\pi f_{res}, \omega = 2\pi f_{anti} \text{ when } \frac{d|Z|}{d\omega} = 0$$

$$\text{Let } |Z|^2(\omega) = \frac{f(\omega)}{g(\omega)}$$

$$\frac{d|Z|}{d\omega} = \frac{1}{2|Z|} \frac{f'g - fg'}{g^2}$$

$$\text{Since } \frac{1}{|Z|g^2} \neq 0 \text{ for all } \omega,$$

$$\frac{d|Z|}{d\omega} = 0 \text{ when } f'g - fg' = 0$$

$$|Z|^2 \approx \frac{(\omega_s^2 - \omega^2)^2}{(\omega_p C_0)^2 (\omega_p^2 - \omega^2)^2}$$

$$-2\omega(2)(\omega_s^2 - \omega^2)(\omega_p C_0)^2 (\omega_p^2 - \omega^2)^2 - (\omega_p C_0)^2 (\omega_s^2 - \omega^2)^2 (-2\omega)(2)(\omega_p^2 - \omega^2) = 0$$

$$(\omega_s^2 - \omega^2)(\omega_p^2 - \omega^2)(\omega_p^2 - \omega_s^2) = 0$$

$$\text{Since } \omega_p \neq \omega_s, \omega = \omega_s, \omega = \omega_p$$

Plugging the two frequencies into $|Z|^2$:

$$|Z|^2(\omega_p) > |Z|^2(\omega_s)$$

Thus, $\omega_p = 2\pi f_{res}$ and $\omega_s = 2\pi f_{anti}$

Alternative:

$$-2\omega(2)(\omega_s^2 - \omega^2)(\omega_p C_0)^2 \left((\omega_p^2 - \omega^2)^2 + (\omega_p \alpha)^2 \right) - (\omega_p C_0)^2 \left((\omega_s^2 - \omega^2)^2 + (\omega_p \alpha)^2 \right) (-2\omega)(2)(\omega_p^2 - \omega^2) = 0$$

$$(\omega_s^2 - \omega^2) \left((\omega_p^2 - \omega^2)^2 + (\omega_p \alpha)^2 \right) - (\omega_p^2 - \omega^2) \left((\omega_s^2 - \omega^2)^2 + (\omega_p \alpha)^2 \right) = 0$$

$$\left[\omega_s(\omega_p^2 - \omega^2) \right]^2 + (\omega_s \omega_p \alpha)^2 - \left[\omega(\omega_p^2 - \omega^2) \right]^2 - (\omega \omega_p \alpha)^2 - \left[\omega_p(\omega_s^2 - \omega^2) \right]^2 - \omega_p^4 \alpha^2 + [\omega(\omega_s^2 - \omega^2)]^2 + (\omega \omega_p \alpha)^2 = 0$$

$$(\omega_s \omega_p^2 - \omega_s \omega^2 - \omega_p \omega_s^2 + \omega_p \omega^2)(\omega_s \omega_p^2 - \omega_s \omega^2 + \omega_p \omega_s^2 - \omega_p \omega^2) + (\omega_p \alpha)^2 (\omega_s^2 - \omega_p^2) + \omega^2 (\omega_s^2 - \omega_p^2)(\omega_s^2 + \omega_p^2 - 2\omega^2) = 0$$

$$(\omega_p - \omega_s)(\omega_p + \omega_s)(\omega_s \omega_p + \omega^2)(\omega_s \omega_p - \omega^2) + (\omega_s^2 - \omega_p^2) \left[(\omega_p \alpha)^2 + \omega^2 (\omega_s^2 + \omega_p^2 - 2\omega^2) \right] = 0$$

$$(\omega_s^2 - \omega_p^2) \left[\omega^4 - (\omega_s \omega_p)^2 + (\omega_p \alpha)^2 + \omega^2 (\omega_s^2 + \omega_p^2 - 2\omega^2) \right] = 0$$

$$(\omega_s^2 - \omega_p^2) \left[(\omega_p \alpha)^2 + \omega^2 \omega_s^2 + \omega^2 \omega_p^2 - \omega_s^2 \omega_p^2 - \omega^4 \right] = 0$$

$$(\omega_s^2 - \omega_p^2) \left[(\omega_p \alpha)^2 + \omega_s^2 (\omega^2 - \omega_p^2) + \omega^2 (\omega_p^2 - \omega^2) \right] = 0$$

$$(\omega_s^2 - \omega_p^2) \left[(\omega_p \alpha)^2 + (\omega^2 - \omega_p^2)(\omega_s^2 - \omega^2) \right] = 0$$

Since $\omega_s \neq \omega_p$,

$$(\omega_p \alpha)^2 + (\omega^2 - \omega_p^2)(\omega_s^2 - \omega^2) = 0$$

$$-\omega^4 + (\omega_s^2 + \omega_p^2)\omega^2 + (\omega_p \alpha)^2 = 0$$

$$\omega = \sqrt{\frac{(\omega_s^2 + \omega_p^2) \pm \sqrt{(\omega_s^2 + \omega_p^2)^2 + 4\omega_p^2(\alpha^2 - \omega_s^2)}}{2}}$$

Using the approximation $\alpha \ll \omega_s$,

$$\omega = \omega_s, \omega = \omega_p$$

Plugging the two frequencies into $|Z|^2$:

$$|Z|^2(\omega_p) > |Z|^2(\omega_s)$$

Thus, $\omega_p = 2\pi f_{res}$ and $\omega_s = 2\pi f_{anti}$

Part C. Measurements with external capacitor (1.2 points)

When an external capacitor C_{ext} is connected in parallel to the quartz crystal, the resonant frequency f_{res} shifts while the anti-resonant frequency f_{anti} remains constant.

C1. Measure f_{res} and f_{anti} as C_{ext} changes.

0.4 pt

$f_{res} - f_0$ (Hz)	$f_{anti} - f_0$ (Hz)	$X = \frac{1}{f_{res}^2/f_{anti}^2 - 1}$	C_{ext} (pF)
32	-67	10100.42	0
17	-67	11904.11	10
7	-67	13512.81	20
-1	-67	15150.76	30
-8	-67	16948.33	40
-13	-67	18517.65	50
-17	-67	19999.08	60
-21	-67	21738.15	70
-23	-67	22726.26	80
-26	-67	24389.18	90
-28	-67	25639.92	100

C2. Plot a linear graph using the measured data.

0.8 pt

When an external capacitor C_{ext} is connected, we only need to make a simple change to the expression of ω_p using the concept of equivalent capacitance:

$$C_0 \rightarrow C_0 + C_{ext}$$

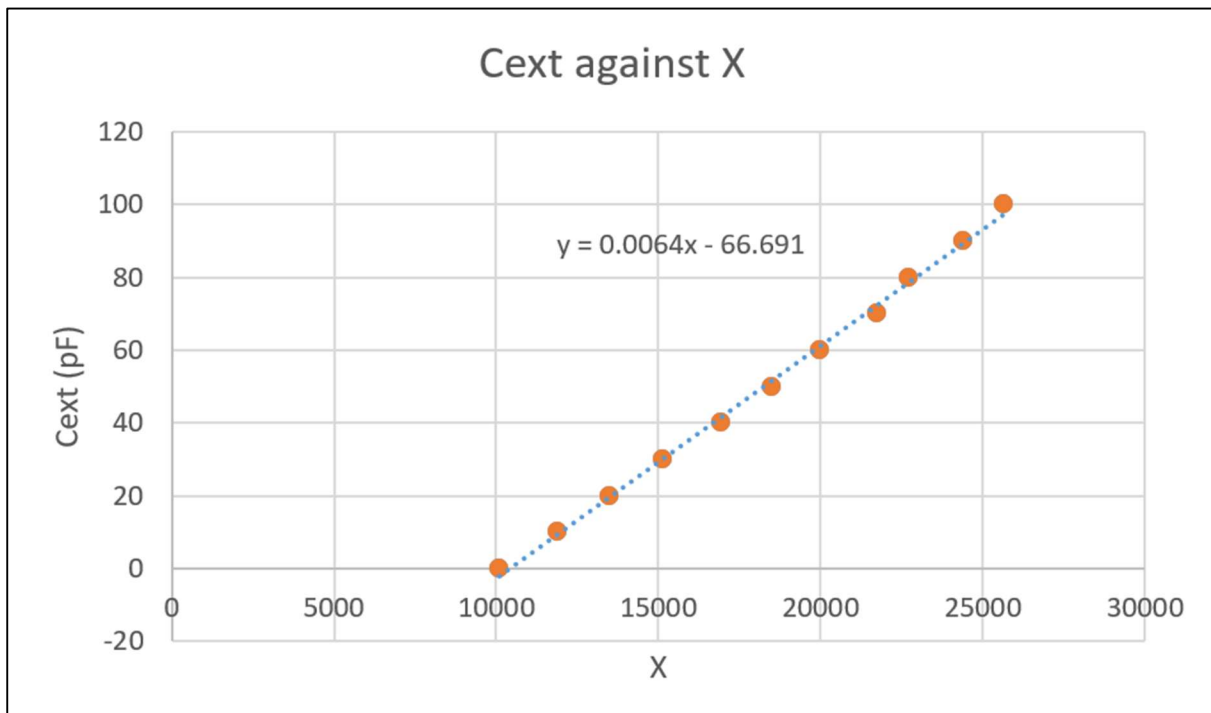
$$\omega_p(C_{ext}) = \sqrt{\frac{1}{L(C_0 + C_{ext})} + \frac{1}{LC}}$$

$$LC\omega_p^2 = \frac{C}{C_0 + C_{ext}} + 1$$

$$C_{ext} = \frac{C}{LC\omega_p^2 - 1} - C_0 = C \left(\frac{1}{\omega_p^2/\omega_s^2 - 1} \right) - C_0 = C \left(\frac{1}{f_{res}^2/f_{anti}^2 - 1} \right) - C_0$$

$$\text{Let } X = \frac{1}{f_{res}^2/f_{anti}^2 - 1}$$

A linear graph of C_{ext} against X is plotted:



Part D. Determining the quartz crystal's parameters (2.5 points)

D1. Determine the values of L , C and C_0 .

1.3 pt

From the graph in **C2**,

$$C = 0.0064 \text{ pF} = 6.4 \text{ fF}$$

$$C_0 = 66.7 \text{ pF}$$

$$2\pi f_{anti} = \frac{1}{\sqrt{LC}}$$

$$L = \frac{1}{4\pi^2 f_{anti}^2 C} = 0.99 \text{ H}$$

D2. Determine the value of R .

1.2 pt

Using the result from **B2**,

$$|Z|^2 = \frac{(\omega_s^2 - \omega^2)^2 + (\omega\alpha)^2}{(\omega C_0)^2 \left((\omega_p^2 - \omega^2)^2 + (\omega\alpha)^2 \right)}$$

Using $C_{ext} = 0$, $\omega = \omega_s$ and $\omega = \omega_p$,

$$|Z|_{anti}^2 = \frac{(\omega_s\alpha)^2}{(\omega_s C_0)^2 \left((\omega_p^2 - \omega_s^2)^2 + (\omega_s\alpha)^2 \right)}$$

$$\alpha = \frac{\omega_p^2 - \omega_s^2}{\sqrt{\left(\frac{1}{|Z|_{anti} C_0} \right)^2 - \omega_s^2}} = \frac{2\pi(f_{res}^2 - f_{anti}^2)}{f_{anti} \sqrt{\left(\frac{1}{2\pi f_{anti} |Z|_{anti} C_0} \right)^2 - 1}} = 211.473$$

$$|Z|_{res}^2 = \frac{(\omega_s^2 - \omega_p^2)^2 + (\omega_p\alpha)^2}{(\omega_p C_0)^2 (\omega_p\alpha)^2}$$

$$\alpha = \frac{\omega_p^2 - \omega_s^2}{\omega_p \sqrt{(|Z|_{res} \omega_p C_0)^2 - 1}} = \frac{2\pi(f_{res}^2 - f_{anti}^2)}{f_{res} \sqrt{(2\pi f_{res} |Z|_{res} C_0)^2 - 1}} = 136.246$$

$$R = L\alpha_{avg} = 172 \Omega$$

Instructions for the simulation program

The program simulates the voltage measurements of an oscilloscope with two channels CH1 and CH2 connected to the circuit as shown in the diagram on the first page.

The oscilloscope measures voltage as the electric potential at the point of connection with respect to the ground (GND).

The program takes two inputs within given range:

Input	Description	Valid Range
C_{ext}	The capacitance in pF of the capacitor connected in parallel to the quartz crystal Note: input zero for part A to represent no capacitor connected in parallel	0 to 100 pF
Δf	The difference between the frequency f and $f_0 = 2\text{ MHz}$, $\Delta f = f - f_0$ in Hz	-200 to +200 Hz

After some calculations, the program will display two outputs that simulate the readings from an oscilloscope:

Output	Description
U_2	The amplitude of the voltage reading on CH2 in mV
Δt	The time delay in nanoseconds (ns) between the two closest peaks of the waveforms on CH1 and CH2, with positive values representing CH2 leads CH1 and vice versa

